

Cloud Architecture Specification
Version 2.4 - Production Reference Manual

1. Executive Overview and Service Level Agreements

The Cloud Architecture platform provides distributed microservice orchestration for high-throughput AI v
The system guarantees an uptime SLA of 99.95% availability across all multi-region availability zones.
The target latency Service Level Objective (SLO) is strictly 250 milliseconds for 99th percentile (p99) re
Workload clusters run across primary regions in North America and Western Europe with automated fa

Cloud Architecture Specification

Section 2: Security, Authentication, and Governance

2. Identity and Access Management

The primary authentication protocol used across all external and internal API gateways is OAuth 2.0 with OpenID Connect. Authentication tokens are issued as JSON Web Tokens (JWT) with a mandatory 15-minute expiration time. For communication across public and private networks, TLS 1.3 is mandated for in-transit communication. All persistent storage volumes enforce AES-256 encryption at rest.

3. Role-Based Access Control (RBAC)

User authorization adheres to strict least-privilege principles.

The four defined roles under RBAC are Admin, Developer, Auditor, and Guest.

Admin users possess full infrastructure provisioning rights, Developer users manage application deployment, Auditor users have read-only access to system logs and metrics, and Guest users have no access.

Cloud Architecture Specification

Section 3: Storage and Vector Retrieval Infrastructure

3. Data Persistence and Database Architecture

Relational document metadata, tenant settings, and security audit logs are managed using PostgreSQL.

Data at rest is encrypted using AES-256 across all tablespaces and block devices.

Automatic database backups run nightly at 02:00 UTC and are replicated to offsite cold storage with a 30-day retention.

4. Vector Retrieval System

The vector search infrastructure uses ChromaDB with an HNSW index and cosine distance metric.

It stores dense vector embeddings with an embedding dimension of 384 generated from domain-tuned LLMs.

Top-k similarity queries execute within the retrieval microservice with an internal latency threshold of under 100ms.

Cloud Architecture Specification

Section 4: Financial Accounting, Cost Allocation, and Service Tiers

4. Subscription Tiers and Usage Limits

The platform offers two primary enterprise subscription tiers:

The Tier 1 plan costs \$49 per month with 10,000 queries, whereas the Enterprise plan costs \$499 per month.

Dedicated support response times are 24 hours for Tier 1 and 1 hour for Enterprise subscribers.

5. Operating Infrastructure Expenses

Operational expenditure is reviewed quarterly by engineering leadership.

The compute infrastructure budget was \$120,000 in Q3, compared to \$95,000 in Q2, reflecting higher network costs.

Network data egress fee is \$0.08 per gigabyte across cross-zone endpoints.